

We shall now go through a few CAT vulnerabilities and look at how to use Ansible to protect against them.

RHEL-06-000005

The rule version (STIG-ID) RHEL-06-000005 states that the audit system must send an email to the designated staff members when the storage volume reaches its capacity. The 'space_left_action' variable in `/etc/audit/auditd.conf` should be set to an email address of an administrator. The following Ansible code snippet checks the same, and asserts if an '@' symbol for the email address exists.

```
---
- name: STIG
  hosts: centos
  become: true
  gather_facts: true
  tags: [STIG]

  tasks:
    - name: "RHEL-06-000005 | CAT II | The audit system must alert designated staff members"
      shell: "grep ^space_left_action /etc/audit/auditd.conf"
      register: space_left_action_result

    - assert:
        that:
          - "'@' in space_left_action_result.stdout"
```

RHEL-06-000008

This rule states that the Red Hat GPG keys need to be installed in order to cryptographically verify that the packages are actually from Red Hat. The 'gpg-pubkey' software package should be installed in the system.

```
- name: "RHEL-06-000008 | CAT I | Vendor-provided cryptographic certificates must be installed"
```

```
shell: "rpm -q gpg-pubkey"
register: gpg_pubkey_result
```

```
- assert:
  that:
    - "gpg_pubkey_result.stdout != ''"
```

RHEL-06-000011

The system should always be up to date with the latest software. This can be checked from the output of 'yum check-update'. The remedy is to run 'yum update' to pull the latest software packages from the repositories, as follows:

```
- name: "RHEL-06-000011 | CAT II | System security patches and updates must be installed"
```

```
shell: "yum check-update"
register: yum_check_update_result
```

```
- assert:
  that:
    - "yum_check_update_result.rc == 0"
```

RHEL-06-000013

This rule states that the downloaded packages need to be cryptographically verified before proceeding with the installation. The 'gpgcheck=1' verification should exist in the `/etc/yum.conf` file.

```
- name: "RHEL-06-000013 | CAT II | System package management tool must be cryptographically verified"
  shell: "grep gpgcheck /etc/yum.conf"
  register: gpgcheck_result
```

```
- assert:
  that:
    - "'=1' in gpgcheck_result.stdout"
```

RHEL-06-000016

The system should have a file integrity tool installed. Advanced Intrusion Detection Environment (AIDE) is a file and directory integrity checker, released under the GPL. It can be installed on CentOS, as follows:

```
- name: "RHEL-06-000016 | CAT II | A file integrity tool must be installed"
  shell: "rpm -q aide"
  register: aide_result
```

```
- assert:
  that:
    - "aide_result.stdout != ''"
```

RHEL-06-000018

The AIDE file integrity tool must be initialised before using it. An initial database needs to be created for its use. The database consists of regular expression rules obtained from the configuration files. A number of digest algorithms such as md5, sha1, sha256, sha512, crc32, etc, are used to check the integrity of files.

```
- name: "RHEL-06-000018 | CAT II | A file integrity baseline must be created"
```

```
stat:
  path: /var/lib/aide/aide.db.gz
  register: aide_db_result
```

```
- assert:
  that:
```